

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Computer Vision – 21CSE454T

UNIT 2: SHAPES AND REGIONS | SESSION 5: ACTIVE CONTOURS

SLO 1: Explain the concept and working principle of active contours for object boundary detection in images.

PROCESS ARRANGEMENT SOLUTIONS

Correct Order for Active Contour Operation:

1. Initialize a contour near the object.
2. Energy is minimized using internal and external forces.
3. The contour gradually moves toward the object boundary.
4. The contour stabilizes at the object boundary.

ACTIVITY 4 – CONCEPT MAPPING SOLUTIONS

Term	Matched Description
Active Contour	B. Boundary detection technique
Snake Model	A. Initial curve placed near the object
External Force	C. Pulls contour toward image edges
Internal Force	D. Maintains smoothness of contour

ACTIVITY 5 – THINK AND APPLY SOLUTIONS

Scenario: Detecting the boundary of a cell in a medical image.

1. Why is active contour useful for this task?

Active contours adapt dynamically to irregular, smooth, and complex cell boundaries. They bridge weak edge gradients and handle low contrast or noisy environments typical of microscopic medical images.

2. What might happen if the initial contour is placed far from the object?

If initialized too far away, external forces (image gradients) become weak, causing the algorithm to converge onto incorrect local minima, lock onto neighboring background noise, or fail to deform toward the target cell entirely.